

B.Sc. (HONS.) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020

STATISTICS AND PROBABILITY

[According to the New Syllabus]

Subject Code : CSE-510227

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | | Marks | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|-----|---|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|----|----|-----|-----|----|-----|----|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|----|----|-----|-----|-----|-----|--|
| 1. | (a) Define statistics, population, sample. | 3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (b) Describe the characteristics of statistics. | 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (c) Distinguish between primary and secondary data. | 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (d) With a suitable example briefly explain the various methods of scales of measurement. | 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2. | (a) What do mean by frequency distribution? Explain the different types of frequency curve. | 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (b) What is variable? Explain various kinds of variable with example. | 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (c) The daily wages taka of 40 workers are given below : | 10 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | <table border="0"> <tr> <td>88</td><td>203</td><td>108</td><td>68</td><td>101</td><td>129</td><td>119</td><td>149</td><td>210</td><td>103</td> </tr> <tr> <td>96</td><td>104</td><td>146</td><td>127</td><td>97</td><td>89</td><td>123</td><td>108</td><td>94</td><td>149</td> </tr> <tr> <td>93</td><td>118</td><td>136</td><td>187</td><td>102</td><td>191</td><td>148</td><td>163</td><td>172</td><td>93</td> </tr> <tr> <td>104</td><td>132</td><td>131</td><td>142</td><td>92</td><td>87</td><td>144</td><td>183</td><td>105</td><td>195</td> </tr> </table> | 88 | 203 | 108 | 68 | 101 | 129 | 119 | 149 | 210 | 103 | 96 | 104 | 146 | 127 | 97 | 89 | 123 | 108 | 94 | 149 | 93 | 118 | 136 | 187 | 102 | 191 | 148 | 163 | 172 | 93 | 104 | 132 | 131 | 142 | 92 | 87 | 144 | 183 | 105 | 195 | |
| 88 | 203 | 108 | 68 | 101 | 129 | 119 | 149 | 210 | 103 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 96 | 104 | 146 | 127 | 97 | 89 | 123 | 108 | 94 | 149 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 93 | 118 | 136 | 187 | 102 | 191 | 148 | 163 | 172 | 93 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 104 | 132 | 131 | 142 | 92 | 87 | 144 | 183 | 105 | 195 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (i) Construct a continuous frequency distribution table with suitable size of class interval. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (ii) Draw a frequency curve and polygon. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (iii) Estimate the limits of wages of central 50% of the workers. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3. | (a) Define and explain central tendency. | 3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (b) What are the essential characteristics of an ideal average? Which measure of central tendency is the best and why? | 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (c) Prove that for two non-zero positive values $AM \geq GM \geq HM$. | 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | (d) The average income of a male employee in a firm is Tk. 1,000 and that of female is Tk. 800. But the mean income of employee is Tk. 920. Find the percentage of male and female employee. | 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

[Please turn over

- Marks
4. (a) Distinguish between absolute and relative measures of dispersion. 5
- (b) The amounts of blood pressure of a patients seen for a sample of 5 days last year were : 103, 97, 101, 106 and 103. Determine the mean deviation. 5
- (c) What is skewness? Explain different kinds of skewness with the help of graph. 6
- (d) A and B are two factory. The average weekly wages and standard deviation of wages of the workers are given below : 4

Factory	Average weekly Wages (Tk.)	Standard deviation	Number of workers
A	1560	90	200
B	1580	70	160

Calculate the combined mean and standard deviation of wages of the whole workers in the two factory.

5. (a) Distinguish between central moments and raw moments. 5
- (b) Briefly express the relationship between the raw moments and central moments. 7
- (c) What is rank correlation? Why we need rank correlation? 3
- (d) Calculate the rank correlation co-efficient from the following data on hourly sales (X) and expenses (Y) of 10 stores. 5

X	50	50	55	60	65	65	65	60	60	60
Y	11	13	14	16	12	19	20	15	17	18

6. (a) Distinguish between correlation and regression. 5
- (b) Regression coefficient is not a pure number—Explain. 5
- (c) How do you estimate the regression line? 5
- (d) By using the following data : 5

X	30	45	32	28	27	24
Y	28	40	27	24	26	18

Estimate the regression equation of Y on X.